

//Exercise 1: Control Structures

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

Question: Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

```
BEGIN

FOR cust_rec IN (

    SELECT CustomerID

    FROM Customers

    WHERE Age > 60

) LOOP

    UPDATE Loans

    SET InterestRate = InterestRate - 1

    WHERE CustomerID = cust_rec.CustomerID;

END LOOP;

COMMIT;

END;
```

Query result Script output DBMS output Explain Plan SQL history				
  Download ▾ Execution time: 0.006 seconds				
	LOANID	CUSTOMERID	INTERESTRATE	DUE DATE
1	101	1	7	7/12/2026, 1:36:20
2	102	2	12	8/1/2026, 1:36:20 P
3	103	3	8	7/7/2026, 1:36:20 P
4	104	4	9	7/17/2026, 1:36:20

Scenario 2: A customer can be promoted to VIP status based on their balance.

Question: Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$10,000.

```
BEGIN

  FOR cust_rec IN (

    SELECT CustomerID

    FROM Customers

    WHERE Balance > 10000

  ) LOOP

    UPDATE Customers

    SET IsVIP = 'TRUE'

    WHERE CustomerID = cust_rec.CustomerID;

  END LOOP;

COMMIT;

END;
```

Query result

Script output

DBMS output

Explain Plan

SQL history

Download

Execution time: 0.008 seconds

	CUSTOMERID	NAME	AGE	BALANCE	ISVIP	
1	1	John	65	15000	TRUE	
2	2	Alice	45	8000	FALSE	
3	3	Bob	70	12000	TRUE	
4	4	Emma		30	20000	TRUE

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

Question: Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

```
SET SERVEROUTPUT ON;
```

```
BEGIN
```

```
  FOR loan_rec IN (
```

```
    SELECT c.Name,
```

```
           l.LoanID,
```

```
           l.DueDate
```

```
  FROM Customers c
```

```
  JOIN Loans l
```

```
  ON c.CustomerID = l.CustomerID
```

```
  WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30
```

```
) LOOP
```

```
  DBMS_OUTPUT.PUT_LINE(
```

```
    'Reminder: Dear ' || loan_rec.Name ||
```

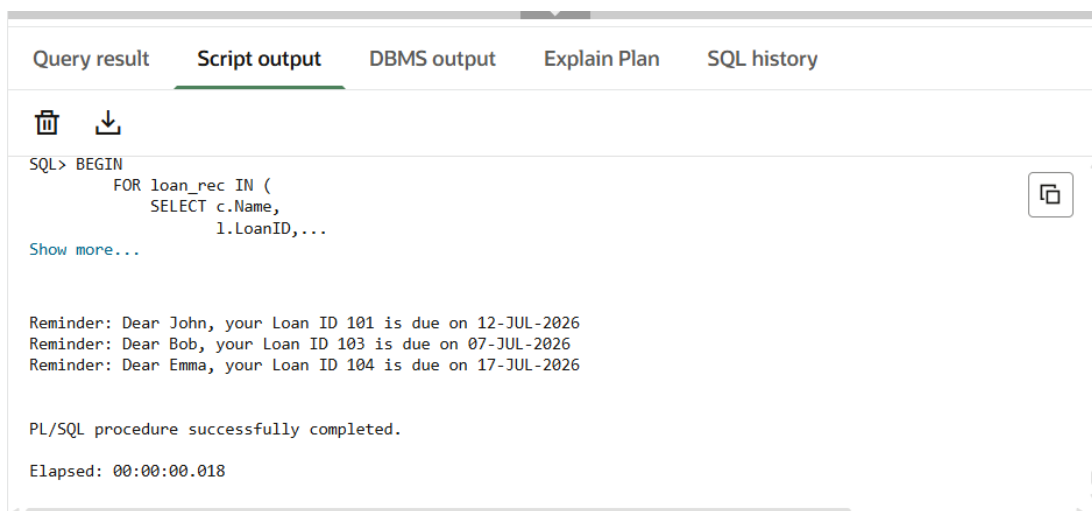
```
    ', your Loan ID ' || loan_rec.LoanID ||
```

```
    ' is due on ' || TO_CHAR(loan_rec.DueDate, 'DD-MON-YYYY')
```

```
  );
```

```
END LOOP;
```

```
END;
```



```
SQL> BEGIN
      FOR loan_rec IN (
        SELECT c.Name,
               l.LoanID,...
      ) LOOP
        DBMS_OUTPUT.PUT_LINE(
          'Reminder: Dear ' || loan_rec.Name ||
          ', your Loan ID ' || loan_rec.LoanID ||
          ' is due on ' || TO_CHAR(loan_rec.DueDate, 'DD-MON-YYYY')
        );
      END LOOP;
    END;
```

Reminder: Dear John, your Loan ID 101 is due on 12-JUL-2026
Reminder: Dear Bob, your Loan ID 103 is due on 07-JUL-2026
Reminder: Dear Emma, your Loan ID 104 is due on 17-JUL-2026

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.018